Input-Aware Control Barrier Function for the Target-Visibility Guarantee

A High-Order CBF alternative to the funnel-margin cone clamp of VDF-ASMC

Soft–Precise–Landing

5 June 2026

Scope. This note develops the target-visibility guarantee as a High-Order Control Barrier Function (HOCBF) on the VDF-ASMC image-kinematic model, in the manuscript's notation — the theory (§1–§8) followed by the per-cycle implementation steps (§9), each step linked to the assumption it rests on. The guarantee is stated on the image *features* $(\boldsymbol{s}, \boldsymbol{h}, \boldsymbol{w}, \alpha)$: the perception front-end that produces them passes neither the markers' physical size nor their shape across the control boundary — only the features enter the law — so the CBF is depth-, scale-, and shape-free at deployment.

Notation

Symbol	Meaning
$\mathcal{I},\ \mathcal{B},\ \mathcal{C},\ \mathcal{V}$	inertial, body, camera, virtual frames
${}^{\mathcal{C}}\hat{\boldsymbol{r}}_i,\ {}^{\mathcal{V}}\hat{\boldsymbol{r}}_i$	i-th image feature point on camera / virtual plane, $i \in \{1,\ldots,N\}$
${}^{\mathcal{V}}\hat{\boldsymbol{r}}$	virtual image point (centroid), ${}^{\mathcal{V}}\hat{\boldsymbol{r}} = N^{-1}\sum_i {}^{\mathcal{V}}\hat{\boldsymbol{r}}_i$
$\boldsymbol{s} = [{}^{\mathcal{V}}\hat{\boldsymbol{r}};\ 1]$	virtual image position (normalized, rad), $\boldsymbol{s}_\mathrm{d} = [0,0,1]^\top$
$\boldsymbol{h},\ \boldsymbol{w}$	translational, rotational optic flow ($\mathcal{V}$-frame), rad/s; h_z = divergence
L_s	stacked image interaction matrix, ${}^{\mathcal{V}}\dot{\hat{\boldsymbol{r}}}_i = L_{s,i}[\boldsymbol{h}\,;\boldsymbol{w}]$
${}^{\mathcal{I}}\boldsymbol{a}_\mathrm{d},\ {}^{\mathcal{V}}\boldsymbol{a}_\mathrm{d}$	inertial / virtual-frame commanded acceleration, ${}^{\mathcal{V}}\boldsymbol{a}_\mathrm{d} = {}^{\mathcal{I}}R_{\mathcal{V}}^\top\,{}^{\mathcal{I}}\boldsymbol{a}_\mathrm{d}$
$\beta = 1/{}^{\mathcal{V}}z_\mathrm{t}$	depth scale (unobservable from the monocular image)
$\mathcal{R} = [r_h, r_w]^\top,\ f$	camera resolution, focal length (px)
$\boldsymbol{\varphi}_\mathrm{max} = \mathcal{R}/(2f)$	camera half-FoV in feature (tangent) units; $\lvert\bar{r}_{\mathrm{e}_k}\rvert = 1$ is the FoV edge
$\boldsymbol{\delta}$	feature half-spread (marker image extent), $\delta_k = \frac{1}{2}\,\mathrm{ptp}_k(\cdot)$
$\theta_\mathrm{curr},\ \theta_\mathrm{cone},\ \theta_\mathrm{cap}$	body tilt $\arccos R_{33}$, cone half-angle, cap ($= 60^\circ$)
$d^\mathrm{fov}_\mathrm{min},\ \boldsymbol{p}_1(t)$	funnel margin, target image funnel (manuscript)
$h_b,\ h_{b,k}$	**(new)** visibility barrier; $h_b = \min_k h_{b,k}$
$k_0,\ k_1$	**(new)** HOCBF stiffness, damping
$\tan\theta_k$	**(new)** per-axis tilt control, $\tan\theta_k = {}^{\mathcal{V}}a_{\mathrm{d},k}/{}^{\mathcal{V}}a_{\mathrm{d},z}$
λ	**(new)** divergence–loom scalar $\lambda = \beta\, a_z = \dot{h}_z + h_z^2$ (observable, scale-free)
$A_k,\ b_k$	**(new)** per-axis HOCBF row $A_k \tan\theta_k \le b_k$

1 Goal: from the cone clamp to a forward-invariant barrier

VDF-ASMC enforces visibility with the *funnel-margin cone clamp*: a shrinking target image funnel $\boldsymbol{p}_1(t)$ sizes a state-dependent attitude cone $\theta_\mathrm{cone}(t)$ that projects the commanded lateral acceleration,

$$ {}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d},xy} \leftarrow \min\Big(1,\ \frac{\lvert {}^{\mathcal{I}}a_{\mathrm{d},z}\rvert \tan\theta_\mathrm{cone}(t)}{\lVert {}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d},xy}\rVert + \epsilon_c}\Big)\,{}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d},xy}, \qquad \theta_\mathrm{cone} = \min\big(\theta_\mathrm{curr} + \tan^{-1}(d^\mathrm{fov}_\mathrm{min}/f),\ \theta_\mathrm{cap}\big). \tag{1} $$

This is a *static, magnitude* projection: it caps $\lVert {}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d},xy}\rVert$ from the present margin $d^\mathrm{fov}_\mathrm{min}$, uses no feature *velocity*, and obtains its guarantee only through the *shrinking* envelope $\boldsymbol{p}_1(t)$. Two costs follow — the shrink fires the perception-death floor early (the marker fills $\boldsymbol{p}_1$ while still inside the real FoV), and the magnitude cap throttles recovery-toward-centre as hard as drift-outward.

The CBF keeps the same visibility guarantee but (i) on the *full* FoV $\boldsymbol{\varphi}_\mathrm{max}$ (no shrink, no early death), (ii) using the optic flow as the feature velocity to obtain forward-invariance *with* recovery preserved, and (iii) input-aware, so the guarantee is one the inner loop can honour.

2 Dynamics used by the CBF

The CBF rides on the VDF-ASMC image-kinematic model, *unchanged.* The virtual image point obeys the position kinematics (manuscript Eq. (s dot))

$$\dot{\boldsymbol{s}} = \boldsymbol{h} - \boldsymbol{w} \times \boldsymbol{s} - \big[(\boldsymbol{h} - \boldsymbol{w} \times \boldsymbol{s}) \cdot \hat{\boldsymbol{e}}_3\big]\boldsymbol{s}, \tag{2}$$

so the centroid velocity is the translational optic flow $\boldsymbol{h}$ up to the known rotational/perspective terms in $\boldsymbol{w}, \boldsymbol{s}$. The translational flow then carries the control through the optic-flow dynamics (manuscript Eq. (h dot))

$$\dot{\boldsymbol{h}} = -\beta\, {}^{\mathcal{I}}R_{\mathcal{V}}^{\top}\, {}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d}} + \tilde{\boldsymbol{c}}_h + \boldsymbol{d}_h = -\beta\, {}^{\mathcal{V}}\boldsymbol{a}_{\mathrm{d}} + \tilde{\boldsymbol{c}}_h + \boldsymbol{d}_h, \tag{3}$$

with $\beta = 1/\,{}^{\mathcal{V}}z_{\mathrm{t}}$ the depth scale, $\tilde{\boldsymbol{c}}_h$ the known kinematic coupling, and $\boldsymbol{d}_h$ the bounded disturbance.

Thus the image position has **relative degree two** with respect to the commanded acceleration: ${}^{\mathcal{V}}\boldsymbol{a}_{\mathrm{d}} \to \dot{\boldsymbol{h}} \to \dot{\boldsymbol{s}} \to \boldsymbol{s}$. The CBF state is $(\boldsymbol{s}, \boldsymbol{h})$, a double integrator in the image plane whose input gain is the depth scale β. **No new model is introduced** — (2)–(3) are the controller's existing equations. Crucially, β is never isolated: §4 parameterizes the command by tilt so β appears only inside the observable product $\beta\, a_z$ (§8).

3 Barrier on the full field of view

Let δ_k be the feature half-spread (marker image extent, measured as half the corner spread). The worst-case feature point lies at $|\,{}^{\mathcal{V}}\hat{r}_k| + \delta_k$ from the optical axis on axis k. Define the per-axis barrier as the slack to the *full* FoV edge $\varphi_{\max,k}$:

$$h_{b,k} = \varphi_{\max,k} - |\,{}^{\mathcal{V}}\hat{r}_k| - \delta_k, \qquad h_b = \min_{k\in\{1,2\}} h_{b,k}, \qquad \mathcal{C} = \{\, h_b \geq 0 \,\}. \tag{4}$$

$h_b \geq 0 \iff$ the outermost feature point is inside the FoV on both axes, i.e. the target (marker) is visible. This is the manuscript funnel margin $d_{\min}^{\mathrm{fov}}$ with the shrinking $\boldsymbol{p}_1(t)$ replaced by the fixed full edge $\boldsymbol{\varphi}_{\max}$. All quantities are image (feature) units; δ_k is read off the image, not computed from any metric size.

Multi-marker realization of $\boldsymbol{\delta}$

On the concentric multi-size board the detected marker set changes with altitude — outer markers are *designed* to exit the frame — so the spread over *all* detected corners would jump as markers leave and would size the barrier off markers meant to depart. Instead the centroid is taken over the whole board (smooth, invariant) while $\boldsymbol{\delta}$ is the half-extent of a single *primary* marker $\mathcal{M}^\star$ (the smallest decodable ID, the last to survive):

$$\delta_k = \tfrac{1}{2}\,\mathrm{ptp}_k\big(\mathcal{M}^\star \text{ corners}\big), \qquad k \in \{1, 2\}. \tag{5}$$

Since $\delta_k \propto \beta$ grows as the UAV descends, a single marker spans only a finite altitude band; the board bounds this by *switching* to a smaller marker before δ_k reaches $\varphi_{\max,k}$, so δ_k follows a **sawtooth** — rising within a band, dropping at each switch — and stays clear of the FoV edge until the *smallest* marker fills the frame (the genuine touchdown floor, where $h_b \to 0$ from the loom δ, not lateral drift, and the lateral CBF is correctly infeasible, §6). Each switch jump is safe-direction (δ_k drops $\Rightarrow h_b$ rises), so forward-invariance is preserved; only $\dot{\delta}_k$ must be *reset* across a switch rather than differentiated through the discontinuity.

4 High-Order CBF and its directionality

Differentiate the binding axis k (the active term of the min). With ${}^{\mathcal{V}}\hat{r}_k = s_k$ and $\dot{s}_k \approx h_k$ from (2),

$$\dot{h}_{b,k} = -\operatorname{sign}(s_k)\,\dot{s}_k - \dot{\delta}_k \;\approx\; -\operatorname{sign}(s_k)\,h_k - \dot{\delta}_k. \tag{6}$$

For $\ddot{h}_{b,k}$ the command enters through (3). Rather than the metric ${}^{\mathcal{V}}a_{\mathrm{d},k}$ — which carries the *unmeasurable* gain β — parameterize the lateral command by the **tilt**, ${}^{\mathcal{V}}a_{\mathrm{d},k} = {}^{\mathcal{V}}a_{\mathrm{d},z}\tan\theta_k$ (a lateral acceleration is the thrust tilted), exactly the variable the cone clamp uses. Then

$$\ddot{s}_k \approx \dot{h}_k = -\beta\,{}^{\mathcal{V}}a_{\mathrm{d},k} + \cdots = -(\beta\,a_z)\tan\theta_k + \cdots = -\lambda\,\tan\theta_k + \ldots, \qquad \boxed{\lambda := \beta\,a_z = \dot{h}_z + h_z^2} \tag{7}$$

where λ — the depth scale times the vertical accel — is **observable from the divergence** (h_z and its rate; a standard loom identity, already part of the manuscript's known coupling $\tilde{\boldsymbol{c}}_h$). Hence $\ddot{h}_{b,k} \approx \operatorname{sign}(s_k)\,\lambda\,\tan\theta_k$, and the HOCBF condition

$$\ddot{h}_{b,k} + k_1\,\dot{h}_{b,k} + k_0\,h_{b,k} \;\geq\; 0 \tag{8}$$

is, being affine in $\tan\theta_k$, **one linear inequality per axis**:

$$\boxed{A_k\,\tan\theta_k \;\leq\; b_k, \qquad A_k = -\operatorname{sign}(s_k)\,\lambda, \qquad b_k = k_0 h_{b,k} + k_1 \dot{h}_{b,k}.} \tag{9}$$

The depth scale β has dropped out: A_k carries only the observable λ and the control is the dimensionless tilt $\tan\theta_k$.

Why it is directional (and the cone clamp is not). With $\lambda > 0$ (positive thrust, $a_z > 0$, $\beta > 0$), the row (9) constrains the *signed* tilt $\operatorname{sign}(s_k)\tan\theta_k$ — the tilt component driving the centroid **outward** (the barrier gradient $\partial h_{b,k}/\partial s_k = -\operatorname{sign}(s_k)$). It is a **half-space** (one-sided) bound,

$$\operatorname{sign}(s_k)\tan\theta_k \;\geq\; -\big(k_0 h_{b,k} + k_1 \dot{h}_{b,k}\big)/\lambda,$$

so only the *drift-increasing* tilt is limited; the *recovery* (inward) and *tangential* tilts stay free. As $h_{b,k} \to 0$ the bound tightens to "no outward tilt" — a hard wall at the edge with full inward authority retained. The cone clamp (1) instead bounds the *magnitude* $\|{}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d},xy}\|$ (a two-sided ball), throttling recovery as hard as drift. Directionality is the difference between constraining $\operatorname{sign}(s_k)\tan\theta_k$ (a half-space) and a norm (a ball).

5 Input-aware projection

The tilt is what the inner loop must deliver, and VDF-ASMC already caps it: $\|\tan\boldsymbol{\theta}\| \leq \tan\theta_{\mathrm{cap}}$, $\theta_{\mathrm{cap}} = 60^\circ$ (Property 1, the 2×-hover-thrust margin). The visibility-safe command is the projection of the ASMC-desired tilt $\tan\theta_k^{\mathrm{des}} = {}^{\mathcal{V}}a_{\mathrm{d},k}/{}^{\mathcal{V}}a_{\mathrm{d},z}$ onto the HOCBF half-spaces intersected with that tilt ball:

$$\boxed{\tan\boldsymbol{\theta}^\star = \arg\min_{\boldsymbol{\tau}\in\mathbb{R}^2} \;\|\boldsymbol{\tau} - \tan\boldsymbol{\theta}^{\mathrm{des}}\|^2 \quad \text{s.t.} \quad A_k\tau_k \leq b_k\ (k=1,2), \quad \|\boldsymbol{\tau}\| \leq \tan\theta_{\mathrm{cap}}.} \tag{10}$$

The ball $\|\boldsymbol{\tau}\| \leq \tan\theta_{\mathrm{cap}}$ is the input-aware ingredient: the barrier can never demand a tilt (hence a lateral accel) the inner loop cannot produce, so the forward-invariance it certifies is one the plant can honour, not a paper guarantee the actuation lag would void.

6 Feasibility floor and hand-off

The constraint set of (10) is *empty* iff even the maximum inward tilt cannot satisfy some row:

$$\min_{\|\boldsymbol{\tau}\|\le\tan\theta_{\text{cap}}} A_k\tau_k = -\tan\theta_{\text{cap}}\,|A_k| \;>\; b_k \quad \text{for some } k. \tag{11}$$

Then $h_b < 0$ is unavoidable — the marker is filling the frame faster than any deliverable tilt can recover — which is the true perception-death floor. Control hands off to a marker-free flow estimator (texture-free ring optic flow) for the blind terminal phase. The CBF thus *certifies* exactly when the visibility objective is physically infeasible, instead of failing silently.

7 Assumptions (and the practical challenge each carries)

1. **The commanded tilt is delivered without lag.** *Challenge:* the inner attitude loop has finite bandwidth; the realized tilt lags the command, and a lagged plant can cross the barrier on the timescale of the delay. The input-aware ball bounds the tilt magnitude but not its delay; the manuscript books this as the attitude-tracking residual inside $\boldsymbol{d}_h$.

2. **Image position has clean relative degree two via $\boldsymbol{h}$.** The rotational/perspective terms $\boldsymbol{w}\times\boldsymbol{s}$ in (2) are treated as known feedforward. *Challenge:* they require a reliable $\boldsymbol{w}$ from $L_s^{\dagger}$; near touchdown the pseudo-inverse degrades.

3. **The gain is the observable λ, at first order.** The lateral coupling is taken as $\lambda\tan\theta_k$ with $\lambda = \dot{h}_z + h_z^2$ (7), which (a) identifies the commanded with the realized vertical accel under thrust tracking, and (b) keeps $\tilde{c}_h, \boldsymbol{d}_h, \ddot{\delta}_k$ only through the measured $\dot{h}_{b,k}$. *Challenge:* $\dot{h}_z$ is a divergence derivative, noisy near touchdown; the reduction is exact only near a centred, steady descent.

4. **The marker is detected** (${}^{\mathcal{V}}\hat{\boldsymbol{r}}_i, \boldsymbol{s}, \boldsymbol{\delta}$ available). *Challenge:* when detection dies the CBF cannot be evaluated — this is the hand-off of (11), not a CBF guarantee.

5. $\tan\theta_{\text{cap}}$ **is the true deliverable tilt bound** (the 60° cap reserves the $2\times$-hover-thrust margin). *Challenge:* this assumes the thrust channel is itself unsaturated.

8 Scale-freeness and depth-freeness

The barrier h_b (4) and the row's right side b_k (9) are built from $\boldsymbol{\varphi}_{\max}$, ${}^{\mathcal{V}}\hat{\boldsymbol{r}}$, $\boldsymbol{\delta}$, $\boldsymbol{h}$, $\boldsymbol{w}$ — all image (feature) quantities. The control gain $A_k = -\operatorname{sign}(s_k)\lambda$ carries the depth scale, but **only inside the product** $\lambda = \beta\, a_z$, never β alone — and that product is observable from the divergence,

$$\lambda = \beta\, a_z = \dot{h}_z + h_z^2, \tag{12}$$

the vertical optic flow h_z and its rate (the manuscript's $\tilde{\boldsymbol{c}}_h$ already forms it). Hence:

- *No isolated β, no metric size.* Parameterizing the command by the tilt $\tan\theta_k$ leaves β only in $\beta\, a_z$, which the divergence supplies. There is **no** $\hat{\beta} = \delta/(f\,\text{size})$ step — that earlier proxy needed the marker's metric size and was therefore equivalent to a metric-depth estimate; it is removed.

- *Image quantities only.* $\dot{h}_z, h_z$ are optic flow; $\tan\theta_k$ is dimensionless; the deliverable bound is $\tan\theta_{\text{cap}}$; δ_k is a measured image extent. No ${}^{\mathcal{V}}z_{\text{t}}$, altitude, marker size, or metric position appears anywhere in the law.

Verdict. The formulation is scale-free, depth-free, and (at the control boundary) shape-free in the same sense as VDF-ASMC: the depth scale enters only through the divergence-observable $\beta\, a_z$, the control is the dimensionless tilt, and the barrier is built from image features. The single honesty caveat is Assumption 3 — the first-order reduction of $\tilde{c}_h, \boldsymbol{d}_h$ and the use of $\dot{h}_z$ — the same practical reduction the manuscript already makes for its optic-flow law.

9 Implementation steps (per control cycle)

Each cycle maps the ASMC command ${}^{\mathcal{V}}\boldsymbol{a}_{\text{d}}$ and the detected feature set to the visibility-safe ${}^{\mathcal{V}}\boldsymbol{a}_{\text{d}}^{\star}$. The right-hand tag links each step to the assumption (Asm., §7) or section it rests on.

1. *Marker selection.* Board centroid ${}^{\mathcal{V}}\hat{\boldsymbol{r}} = N^{-1}\sum_{i=1}^{N} {}^{\mathcal{V}}\hat{\boldsymbol{r}}_i$; primary marker $\mathcal{M}^{\star}$ = smallest decodable ID, with hysteresis to suppress chatter. [Asm. 4]

2. *Barrier* (per axis $k \in \{1,2\}$). δ_k from (5), then $h_{b,k} = \varphi_{\max,k} - |{}^{\mathcal{V}}\hat{r}_k| - \delta_k$, with $\varphi_{\max,k} = \mathcal{R}_k/(2f)$. [Asm. 4]

3. *Barrier rate* (per axis k). Centroid velocity from the optic flow and known feedforward,

$$\dot{\hat{r}}_k \approx h_k - (\boldsymbol{w}\times\boldsymbol{s})_k, \qquad \dot{\delta}_k = \frac{\delta_k(t) - \delta_k(t-\Delta t)}{\Delta t}\bigg|_{\text{reset 0 on switch}}, \qquad \dot{h}_{b,k} = -\operatorname{sign}({}^{\mathcal{V}}\hat{r}_k)\,\dot{\hat{r}}_k - \dot{\delta}_k.$$

[Asm. 2, 3; §3]

4. *Loom gain.* $\lambda = \dot{h}_z + h_z^2$ from the divergence (h_z) and its rate — the observable $\beta\, a_z$; no marker size. [§8]

5. *HOCBF rows* (per axis k).

$$A_k = -\operatorname{sign}({}^{\mathcal{V}}\hat{r}_k)\,\lambda, \qquad b_k = k_0\, h_{b,k} + k_1\, \dot{h}_{b,k}, \qquad \text{row} \quad A_k \tan\theta_k \le b_k.$$

[Asm. 3]

6. *Deliverable bound.* tilt ball $\|\tan\boldsymbol{\theta}\| \le \tan\theta_{\text{cap}}$, $\theta_{\text{cap}} = 60°$. [Asm. 5]

7. *Input-aware projection.* desired tilt $\tan\theta_k^{\text{des}} = {}^{\mathcal{V}}a_{\text{d},k}/{}^{\mathcal{V}}a_{\text{d},z}$; then $\tan\boldsymbol{\theta}^{\star} = \arg\min_{\boldsymbol{\tau}} \|\boldsymbol{\tau} - \tan\boldsymbol{\theta}^{\text{des}}\|^2$ s.t. $A_k\tau_k \le b_k\,(k{=}1,2)$, $\|\boldsymbol{\tau}\| \le \tan\theta_{\text{cap}}$. Closed form: (i) clamp each τ_k to its outward limit b_k/A_k; (ii) if $\|\boldsymbol{\tau}\| > \tan\theta_{\text{cap}}$, scale to the ball; (iii) re-clamp (i). [Asm. 1]

8. *Feasibility / hand-off.* If $-\tan\theta_{\text{cap}}|A_k| > b_k$ for any k: raise the perception-floor flag and switch the flow source to the texture-free ring estimator. [§6]

9. *Command map.* ${}^{\mathcal{V}}a_{\text{d},k}^{\star} = {}^{\mathcal{V}}a_{\text{d},z}\tan\theta_k^{\star}$, then ${}^{\mathcal{I}}\boldsymbol{a}_{\text{d}}^{\star} = {}^{\mathcal{I}}R_{\mathcal{V}}\,{}^{\mathcal{V}}\boldsymbol{a}_{\text{d}}^{\star}$ to the SO(3) inner loop (existing acceleration floor and low-pass filter follow). [–]